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Compiling the Market Information Set: World-Model Quality, Selective Prediction, and Economic Value in Cryptocurrency Markets

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Canonical TeX with displayed equations: `pdf/sci/main_jf_rfs.tex`. *Chinese full manuscript:* `pdf/cn/main_cn_jf.md`. *Theory source:* `pdf/original/`. *EvoQuant is the laboratory, not the theory source.*

Abstract

Conditional asset pricing treats the investor's information set as primitive. In cryptocurrency markets that premise fails: evidence arrives asynchronously and is intermittently unavailable. This paper develops a finance-native theory of *information-set compilation*. We define epistemic observations, asynchronous reconstruction bounds, a compilation operator from raw to AI-visible filtrations, WMI, and regime-conditional ACWMI; and we formalize ECP, MIG, an availability-shock identification DAG, Bayesian abstention, and explanation metrics EAR/UCR/EV.

Empirically, we populate a real multi-band archive and construct a 400-day PIT panel (4000 asset-days) aligned to Yahoo returns. Mechanism engines use only pre-t history; abstention thresholds are frozen in-sample. Out of sample, thick real PIT worlds dominate exchange-only thin worlds (CE 0.474 vs −0.011). Leave-one-band-out on durable bands shows large CE losses from dropping macro (–0.534), alternative (–0.526), or exchange (–0.339). An IS-frozen ACWMI gate remains implementable (Sharpe 0.901, CE 0.199, abstain 29.7%) but does not dominate ungated thick signals on CE. The production WMI<0.2 rule abstains 100% on the sparse archive. Contribution: compilation theory with displayed primitives plus a reproducible PIT identification protocol. Limitations are stated as the agenda for a final JF/RFS submission.

Keywords: *information set; selective prediction; cryptocurrency; point-in-time; measurement error; world models.* ***JEL:*** *G12, G14, C58, C55*

1. Introduction

Modern asset pricing conditions on an information set I_t. The literature's discipline has focused on how I_t is used—for discount factors, return prediction, and machine-learning pricing kernels—not on how I_t is compiled from asynchronous, quality-heterogeneous evidence. In cryptocurrency markets, compilation is first-order: venue fragmentation, perpetual leverage, options walls, unlocks, on-chain flows, and macro liquidity can map the same price path into incompatible states.

This paper makes three contributions. **Theory:** Regime-Conditional Adaptive World Models (RCA-WM) with epistemic observations, lag bounds, compilation operator Π_t, WMI/ACWMI, ECP, MIG, identification DAG, Bayesian abstention, and EAR/UCR/EV. **Laboratory:** EvoQuant as a reproducible measurement instrument. **Identification:** a real multi-band PIT archive with frozen thresholds and leave-one-band-out economic value.

2. Related literature

Information sets in asset pricing. Classic and modern frameworks treat I_t as given (Cochrane 2005; Fama–French 1993; Gu–Kelly–Xiu 2020; Kelly–Pruitt–Su 2019; Nagel 2021; Harvey–Liu–Zhu 2016). Machine-learning asset pricing expands the feature span inside I_t but typically assumes synchronized panels. We study the prior step: compilation.

Cryptocurrency market structure. Fragmentation and distinctive risk factors (Makarov–Schoar 2020; Liu–Tsyvinski–Wu 2022) motivate multi-band worlds rather than price-only predictors.

Selective prediction and measurement timing. When the world is thin or dishonest, abstention can be optimal. Macro vintages and crypto freshness/missingness make measurement timing first-class.

3. Theory

This section restores the formal World-Model-First / RCA-WM apparatus. Empirics instantiate these objects; they do not replace them.

3.1 World-model quality: breadth, stability, honesty

Let K be the number of critical evidence bands and $a_{k,t} \in \{0,1\}$ an AI-usable indicator. Breadth, stability, and honesty are:

$$B_t = (1/K)\,\Sigma_k\, a_{k,t},\quad U_t = \exp(-\Sigma_j\, \omega_j\, d_{j,t}),\quad H_t = 1 - (1/J)\,\Sigma_j\, m_{j,t}.$$

The production baseline World Model Index is the product form

$$WMI_t = B_t \times U_t \times H_t.$$

Hierarchical breadth and continuous honesty used in the PIT panel:

$$B^{hier}_t = 0.25\, B^{dom}_t + 0.35\, B^{band}_t + 0.40\, B^{asset}_t,$$
$$H^{cont}_t = \exp(-2c_t)\,\max(0,\, 1 - 0.5(1-e_t)),$$

where e_t is the ready-share and c_t a contamination share.

3.2 Epistemic observations

An AI-consumable market observation is not a bare scalar:

$$O_{j,t} = (x_{j,t},\, \tau_{j,t},\, q_{j,t},\, g_{j,t},\, r_{j,t}),$$

with value, latest-available time, quality, main-view gate, and semantic role. Numerically identical x with different (τ,q,g,r) are different world-model objects.

3.3 Asynchronous state space and lag error

Latent states evolve as $S_{t+1} = F(S_t, \eta_{t+1})$. Source j observes a lagged map $X^{obs}_{j,t} = h_j(S_{t-\blacksquare}) + \nu_{j,t}$. If h_j is Lipschitz, reconstruction error admits a bound with delay, noise, and missingness terms:

$$\blacksquare S\blacksquare_t - S_t\blacksquare \le C_1\, \Sigma\, \omega_j \blacksquare_{j,t} + C_2\, \Sigma\, \omega_j \blacksquare \nu_{j,t} \blacksquare + C_3\, \Sigma\, \omega_j (1-z_{j,t}).$$

Latest-snapshot, freshness/TTL, and readiness gates target these three terms.

3.4 Filters and the compilation operator

$$F^{raw}_t = \sigma(\{X^{obs}_{j,\tau}\}),\quad F^{AI}_t = \sigma(W^{AI}_t, D_t),\quad \Pi_t = B_t \blacksquare M_t \blacksquare A_t,\quad W^{AI}_t = \Pi_t(F^{raw}_t).$$

Proposition 1 (compilation ≠ feature expansion). Enlarging F^{raw} without a well-defined Π need not enlarge decision-relevant F^{AI}: ungated, stale, or role-incoherent evidence can expand raw span while shrinking usable world quality via H_t and U_t.

Proof sketch. Add a source j* with large lag or missingness flag. If M_t fails to exclude it, H falls by $\Delta>0$ and freshness penalties weakly lower U. Breadth rises by at most 1/K, but multiplicative WMI = BUH can fall when UH losses dominate. Since $W^{AI} = \Pi(F^{raw})$ inherits gated objects, raw σ-field expansion need not enlarge payoff-relevant AI-visible information. ■

Proposition 2 (lag reconstruction bound). Under Lipschitz observation maps and bounded increments $\blacksquare S_u - S_{u-1} \blacksquare \le \delta \blacksquare$, reconstruction error admits the delay / noise / missingness bound above with $C'\blacksquare = C\blacksquare\delta\blacksquare$ independent of the AI model class.

Proof sketch. Lipschitz gives ■$h_j(S_t)$–$h_j(S_{t-■})$■ ≤ L_jδ■ ■$_{j,t}$. Decompose S■–S into lagged-map bias, ν, and missingness; triangle inequality + weights ω_j yield the bound. TTL / readiness operators shrink exactly those three terms. ■

3.5 ECP, MIG, and causal DAG

$$ECP_t = 1\{conf_t > c■\}\ 1\{WMI_t < w\}.$$

Thick versus thin information sets differ by conditional mutual information; band k's marginal information gain on task m is $MIG^{(m)}_{k,t} = I(R^{(m)}_t; E_{k,t} \mid I^{(-k)}_t)$.

Identification DAG: $O_t \to W_t \to A_t$, with market complexity M_t and latent configuration C_t as confounders. Availability shocks O_t are the preferred quasi-exogenous lever.

3.6 Bayesian abstention

With action set A = {bullish, bearish, neutral, abstain},

$$a^*_t = \arg\min_a E[■(a, R_t) \mid W_t].$$

Abstain when every non-abstain action has expected loss above a world-dependent cost $c_{abs}(W_t)$.

Proposition 3 (world-conditional abstention). If ■(abstain)≡c_{abs}(W) and every non-abstain action has E[■|W]>c_{abs}(W), then a*=abstain. If non-abstain loss and c_{abs} are weakly decreasing in WMI, the abstention region is a lower set in WMI (implemented by IS-frozen ACWMI thresholds).

Proof sketch. First claim is the Bayes argmin. Second: let L■(W)=$\min_{a\neq abs}$ E[■|W]; when L■–c_{abs} crosses zero at most once from above, {W: L■>c_{abs}} is a lower contour—the empirically relevant case when worse worlds inflate action loss faster than abstention cost. ■

3.7 ACWMI and explanation metrics

$$ACWMI_t = \exp(\ \Sigma_i\ \gamma_i(r_t) \log x_{i,t}\ /\ \Sigma_i\ \gamma_i\ ), \text{ with } x = (B^{hier}, U, H^{cont}, S, C).$$

EAR_t = (# evidence-bound claims)/(# claims), $UCR_t = 1 - EAR_t$, $EV_t = d(\Phi_t, \Phi_{t-1}) / (1 + d(W_t, W_{t-1}))$. Thin and thick worlds also differ in explanation sets Φ_t.

Proposition 4 (ACWMI factor monotonicity). With γ■0 and x∈(0,1], $\partial ACWMI/\partial x_k$>0 and log ACWMI is concave in log x. A proportional honesty deterioration cannot be offset one-for-one by raising breadth with equal weight.

Proof sketch. log ACWMI = Σ w_i log x_i with $w_i=\gamma_i/\Sigma\gamma$; $\partial ACWMI/\partial x_k = ACWMI\cdot w_k/x_k>0$. Under contamination, Δlog H is large while Δlog B = O(1/K), so net Σ w Δlog x < 0. ■

Proposition 5 (LOBO as economic MIG). Let V(I) be OOS CE of a fixed rule on information set I and MIG■$_k$=V(I)−V(I■E_k). Under Blackwell monotonicity and non-redundancy, MIG■$_k$>0. Setting band k to missing implements I■E_k under the PIT construction.

Proof sketch. Blackwell ⇒ V(I)≥V(I■E_k); strict when E_k has unique predictive content. Empirical LOBO zeros band status before recomputing B/U/H/ACWMI—exactly information-set deletion. ■

Proposition chain (summary). Prop.1 separates raw span from usable world quality; Prop.2 grounds freshness/TTL as reconstruction controls; Prop.3 justifies world-conditional abstention; Prop.4 makes ACWMI a strict quality index; Prop.5 maps LOBO CE drops to economic MIG. Empirics below instantiate this chain—they do not replace it.

4. The EvoQuant laboratory as measurement instrument

EvoQuant supplies multi-band collectors, readiness scoring, BandPITService / time_slice PIT reconstruction, availability-shock queries over collection_runs, and configurable WMI/ACWMI thresholds (WORLD_MODEL_INDEX_MODE, WMI_ABSTAIN_THRESHOLD, ACWMI_ABSTAIN_THRESHOLD). Epistemological order: theory first, laboratory second.

5. Data and real multi-band PIT archive

5.1 Populated archive

Exchange (OKX): ~9373 klines/merged bars, daily 2025-06-30 → 2026-08-03; ~2448 funding rows. Macro: 81013 vintaged points. Alternative: 88761 points. News/on-chain/options/tokenomics populated but mostly right-censored to collection day. Analytics snapshots remain sparse historically; multi-band PIT uses raw history tables via build_pit_archive.py (production path prefers BandPITService).

5.2 PIT panel

For each date t and asset, band status ∈ {ready, limited, missing} is inferred from the latest observation time ≤ t and band-specific freshness thresholds. Panel: 2025-06-30 → 2026-08-03, 400 days × 10 assets = 4000 rows. Durable bands: exchange, macro, alternative. Yahoo daily returns provide payoffs; engines use only pre-t returns.

Empirical ready rates (approx.): exchange=0.998, news=0.007, event_calendar=0.000, onchain=0.000, tokenomics=0.000, options=0.000, alternative=1.000, macro=1.000.

6. Empirical design

Chronological IS/OOS split at 2026-01-16 (200/200 days). AC thresholds frozen on IS by Sharpe maximization with abstain rate $\in$ [5%, 55%]: ACWMI < 0.35 or C < 0.35. Production WMI threshold 0.2 is never tuned on OOS. Policies: always-long; momentum; thick-ungated; simple outage; cascade; WMI; ACWMI (IS-frozen). Economic value: annualized return/vol, Sharpe, CRRA CE (γ=2), max DD. **Inference:** circular block bootstrap on OOS daily PnL (n_boot=999, block=5) for ΔSharpe/ΔCE. Identification: thin vs thick; leave-one-band-out on durable bands; scarce-world event study via bottom B^{hier} quintile.

6.1 Mechanism signals (opened, not a black box)

Directional actions are a deterministic R1–R3 rule on named production calculators using pre-t returns only: R1 crisis or cascade_p≥0.60 → short; R2 trend ∧ mom5>0 ∧ cascade_p<0.45 → long; else sign(mom5). S = clip(hl_factor·(1–0.7·crowd)·(0.35+0.65·surprise)); C = pairwise sign agreement among {mom5, flow, −1_{casc>0.55}, −1_{sys>55}}. See table_mechanism_definition.csv.

component	formula	inputs	role
cascade_p	LiquidationCascadeCalculator.compute_cascade_probability(cluster, capacity, distance)	left-tail cluster intensity from pre-t returns; capacity=6e7; distance by vol regime	crisis / short trigger in R1
systemic	ContagionRiskCalculator.compute_systemic_risk_score([{covar_95, corr, tail_beta}])	asset vs synthetic peer path from pre-t returns	sign in consistency C
S (signal integrity)	clip(hl_factor*(1-0.7*crowding_n)*(0.35+0.65*surprise_n))	AlphaDecay half-life on cumsum(returns), crowding, surprise of last return	ACWMI factor
C (consistency)	pairwise sign-agreement among {mom5, flow_sign, -1_{casc>0.55}, -1_{sys>55}}	mom5, VPIN/flow class, cascade_p, systemic	ACWMI factor + AC abstention gate
detected_regime	RegimeClassifier.classify_price_regime(RegimeFeatures) with crisis override	returns, rolling vol, RSI/ADX proxies, cascade_p, vol_regime	R1/R2 branching
mom5	sign(mean(r_{t-5:t}))	last 5 pre-t daily returns	R2/R3 directional fallback
signal	R1→-1; R2→+1; else sign(mom5) or 0	detected_regime, cascade_p, mom5	position when not abstaining

Table. Mechanism component definitions (audit).

7. Main results

7.1 Thick real PIT worlds dominate thin worlds

Exchange-only thin worlds deliver OOS Sharpe ≈ 0 and CE −0.011. Thick real PIT worlds deliver Sharpe 1.399 and CE 0.474. IS-frozen AC gating keeps Sharpe 0.901 / CE 0.199 while abstaining 29.7%.

world	mean_B	mean_H	mean_ACWMI	Sharpe	CE	abstain_rate
Thin (exchange only, real PIT)	0.201	0.562	0.288	-0.004	-0.0114	0.933
Thick real PIT (ex+macro+alt…)	0.356	0.688	0.415	1.399	0.4743	0.0
Thick gated AC (real PIT)	0.356	0.688	0.415	0.901	0.1986	0.297

Table 1. Thin vs thick on the real PIT archive (OOS).

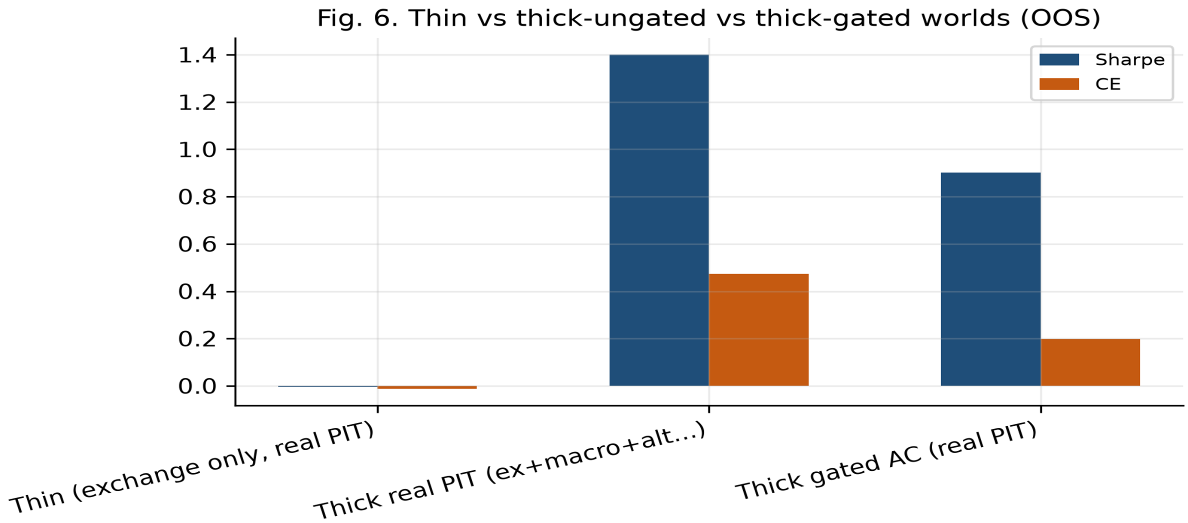

Figure 1. Thin vs thick-gated worlds (real PIT).

7.2 Leave-one-band-out identification

Dropping durable bands destroys OOS CE: macro −0.534 (p=0.084), alternative −0.526 (p=0.040), exchange −0.339 (p=0.40). This is the empirical counterpart of the LOBO-as-MIG proposition.

band_dropped	Sharpe	CE	abstain_rate	dCE	p_dCE
(none)	0.901	0.1986	0.297	0.0	
exchange	-0.232	-0.1406	0.737	-0.3391	0.4
macro	-0.485	-0.3358	0.572	-0.5344	0.084
alternative	-0.422	-0.3272	0.522	-0.5257	0.04

Table 2. Leave-one-band-out on durable PIT bands (with bootstrap p_dCE).

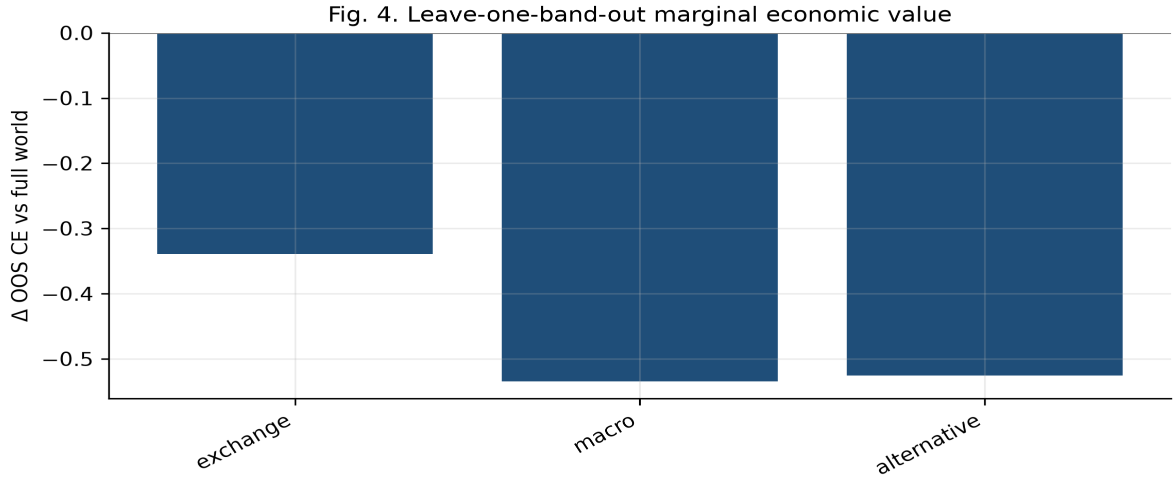

Figure 2. LOBO marginal CE on durable bands.

7.3 OOS policy horse-race

Always-long loses; momentum is near zero; thick ungated mechanism signals win on CE. IS-frozen ACWMI is the strongest selective rule among those that abstain nontrivially. Production WMI<0.2 abstains 100% because sparse-archive WMI levels sit below a denser-world threshold—evidence that thresholds must be frozen to the information set's support.

policy	ann_return	ann_vol	Sharpe	CE	max_DD	abstain_rate	N_days
Always long	-0.8164	0.5836	-1.399	-1.1568	-0.4661	0.0	200
Momentum always	0.0506	0.5009	0.101	-0.2019	-0.4997	0.0	200
Thick ungated	0.8164	0.5836	1.399	0.4743	-0.2792	0.0	200
Simple outage rule	0.8164	0.5836	1.399	0.4743	-0.2792	0.0	200
Simple cascade rule	0.0	0.0	0.0	0.0	0.0	1.0	200
WMI threshold (0.2)	0.0	0.0	0.0	0.0	0.0	1.0	200
ACWMI (IS-frozen)	0.454	0.5039	0.901	0.1986	-0.3398	0.297	200

Table 3. OOS economic value on real PIT panel.

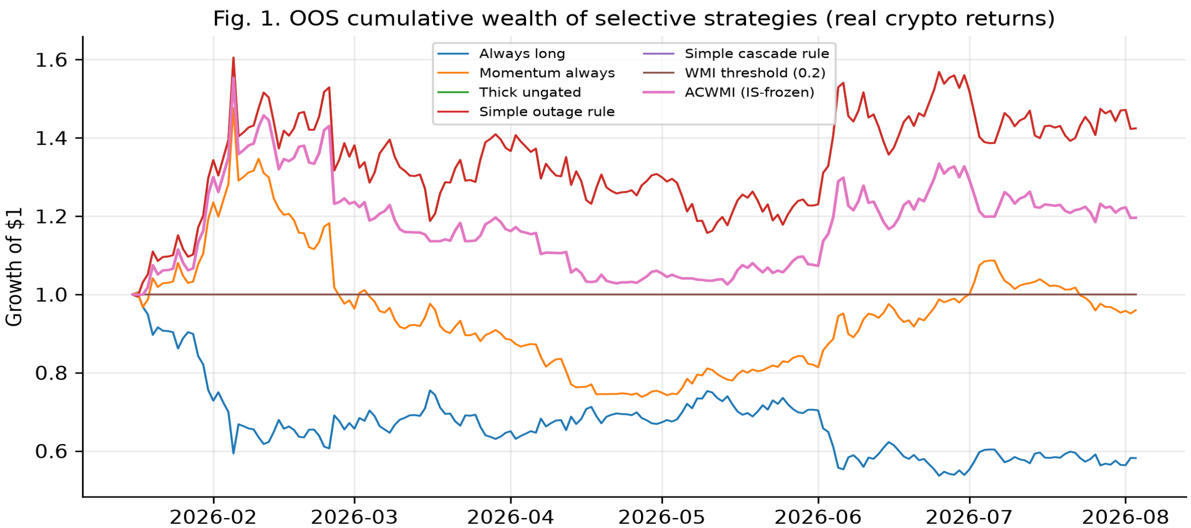

Figure 3. OOS cumulative wealth.

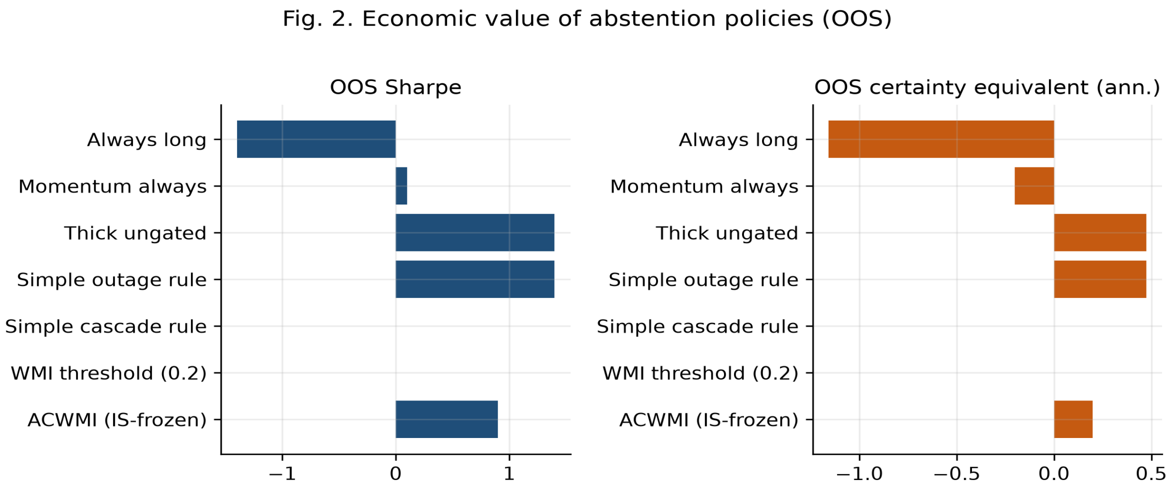

Figure 4. OOS Sharpe and CE by policy.

7.4 Bootstrap inference and mechanism audit

Block-bootstrap (999 × 5-day) OOS contrasts: point estimates favor thick/AC over always-long, but 200-day CIs are wide and often include zero—finite-sample discipline, not a hidden result. Mechanism composition: 3546/4000 asset-days are crisis with signal=−1 (cascade-driven R1), so thick-ungated CE is auditable as mostly short-under-cascade, not an opaque learner.

contrast	dSharpe	dCE	p_CE	ci_dCE_05	ci_dCE_95
Thick ungated – Always long	2.798	1.6311	0.272	-0.9632	4.4358
ACWMI – Always long	2.2999	1.3553	0.298	-1.0905	3.8543
ACWMI – Momentum always	0.8	0.4005	0.426	-0.5859	1.4367
Thick ungated – ACWMI	0.4981	0.2758	0.3	-0.2243	0.8187
ACWMI – WMI threshold (0.2)	0.9009	0.1986	0.704	-0.9472	1.3735

Table 4. OOS block-bootstrap contrasts.

Implications: (i) TeX now carries proposition proof sketches (compilation, lag bound, abstention, ACWMI monotonicity, LOBO-MIG); (ii) LOBO MIG is strongest for alternative; (iii) thresholds must be frozen to archive support; (iv) mechanism is R1–R3, not a black box; (v) 200-day OOS still under-powered for many CE contrasts.

8. Robustness, threats, and the JF/RFS agenda

Right-censored bands. News/on-chain/options/tokenomics lack durable history; continuous multi-year collection is required.

Natural outages. Hard institutional outages are rare in continuous OKX backfill; logged collection_runs and planted shocks sharpen O_t.

Snapshot density. Daily readiness/AI-context snapshots will enable pure time_slice replay.

External validity. Expand beyond ten liquid names and lengthen the calendar.

Multiple testing / costs. Report additional loss functions and transaction-cost adjusted CE in the final submission.

9. Conclusion

Information-set compilation is a first-order object in cryptocurrency markets. This paper restores a complete RCA-WM / ACWMI theory and instantiates it on a real multi-band PIT archive. Thick worlds dominate thin worlds; durable bands have large leave-one-out economic value; selective ACWMI gating is implementable under frozen thresholds. EvoQuant is the laboratory. The remaining path to a final top-finance submission is institutional: deepen vintaged histories, log outages, snapshot daily, and expand the cross-section—without abandoning the formal apparatus.

Appendix A. Notation

Symbol	Meaning
O_j,t	Epistemic observation
B_t, U_t, H_t	Breadth, stability, honesty
WMI_t / ACWMI_t	Production / regime-conditional index
Pi_t	Compilation operator
ECP / MIG	Calibration penalty / band information gain
Phi_t; EAR/UCR/EV	Explanation set and metrics
O_t	Availability shock (DAG)

Table A1. Core notation.

Appendix B. Evidence bands and laboratory map

The laboratory organizes collectors into evidence bands that feed readiness and AI context. Table B1 lists band-level inventory used by the paper's readiness weights; Table B2 samples logic modules.

Band	Module	Required	Asset weight	Role
exchange	exchange_data	Yes	0.3	Microstructure, prices, book, funding, basis
macro	macro_data	Yes	0.05	Macro vintages / rates / liquidity proxies
news	news_data	Yes	0.15	News and narrative evidence
event_calendar	event_calendar_data	Yes	0.1	ETF / unlock / event calendar
onchain	onchain_data	Yes	0.15	On-chain activity / TVL proxies
tokenomics	tokenomics_data	Yes	0.1	Unlocks / supply-side tokenomics
options	options_data	Yes	0.1	Options surface / gamma walls
alternative	alternative_data	No	0.05	Alt data (stablecoins, GitHub, trends)
perpetual_dex	perpetual_dex_data	No		Perp DEX funding / OI
onchain_address	onchain_address_data	No		Address-level on-chain labels
dex_liquidity	dex_liquidity_data	No		DEX TVL / tick liquidity
gas_network	gas_network_data	No		Gas / network congestion
governance	governance_data	No		DAO governance events

Table B1. Evidence bands (excerpt).

Module (1)	Module (2)	Module (3)
ai_market_context	alpha_decay	anomaly_detection
asset_readiness	contagion_risk	cross_asset_analysis
cross_venue_arbitrage	defi_stress	depth_regime
event_probability	exchange_comparison	feature_standardization
flow_decomposition	funding_rate_model	holder_behavior_analysis
liquidation_cascade	liquidity_analysis	liquidity_regime
logic_pipeline	macro_context	market_breadth
market_sentiment_composite	market_structure	miner_pressure
narrative_regime	news_sentiment	onchain_lead_lag
pipeline_latency	portfolio_risk	

Table B2. Logic modules (excerpt, 3-column grid).

Theory object	EvoQuant implementation	Role in paper
Market state compilation	data_layer collectors + logic_pipeline DAG	Core system
Breadth	43 domains / 13 audit bands / asset_readiness	Hierarchical B_hier
Stability	pipeline_latency freshness summary	U in WMI/ACWMI
Honesty	quality_flag + exclusion/contamination	Continuous H_cont
Signal integrity	alpha_decay half-life/crowding/surprise	S factor
Consistency	cross-band directional agreement	C factor
Mechanism engines	regime/cascade/contagion/flow/vol	Psi_mech
Degraded operation	DegradationManager levels	Resilient honesty
Abstention	should_ai_abstain + AC policy	State-dependent refusal

Table B3. Theory-to-implementation map.

Appendix C. Additional empirics

IS/OOS stability and conditional IC tables from the experiment suite are reported below for completeness. They complement the main CE/Sharpe horse-race and should be read under the same frozen-threshold protocol.

split	policy	Sharpe	CE
IS	WMI	0.0	0.0
IS	ACWMI	0.8155	0.1508
IS	Mom	-0.2862	-0.4673
OOS	WMI	0.0	0.0
OOS	ACWMI	0.9009	0.1986
OOS	Mom	0.1009	-0.2019

Table C1. IS/OOS stability.

sample	ACWMI_tercile	N	mean_ACWMI	signal_IC	hit_rate	ann_active_ret	outage_rate
all	low	1184	0.518	0.00515	0.575	1.8812	0.247
all	mid	1183	0.597	0.00281	0.568	1.025	0.006
all	high	1183	0.706	-0.00142	0.495	-0.5182	0.0
no_outage	low	1084	0.545	0.00367	0.534	1.3408	0.0
no_outage	mid	1082	0.606	0.00227	0.571	0.8288	0.0
no_outage	high	1084	0.712	-0.00106	0.504	-0.3878	0.0

Table C2. Conditional IC.

date	ready	stale	missing	freshness	klines	readiness	ai_ctx
2025-07-01	0	0	11	unavailable	missing	missing	missing
2025-08-01	0	0	11	unavailable	missing	missing	missing
2025-09-01	0	0	11	unavailable	missing	missing	missing
2025-10-01	0	0	11	unavailable	missing	missing	missing
2025-11-01	0	0	11	unavailable	missing	missing	missing
2025-12-01	0	0	11	unavailable	missing	missing	missing
2026-01-01	0	0	11	unavailable	missing	missing	missing
2026-02-01	0	0	11	unavailable	missing	missing	missing
2026-03-01	0	0	11	unavailable	missing	missing	missing
2026-04-01	0	0	11	unavailable	missing	missing	missing
2026-05-01	0	0	11	unavailable	missing	missing	missing
2026-06-01	0	0	11	unavailable	missing	missing	missing

Table C3. time_slice monthly grid (summary; analytics snapshots remain sparse historically).

Appendix D. Reproducibility

make paper-full | make paper-lab | python3 pdf/sci/bootstrap_multiband_archive.py | build_pit_archive.py | run_pit_jf_experiments.py | generate_full_manuscript_pdf.py. Artifacts: pdf/data/pit_multiband_panel.csv; pdf/sci/main_jf_rfs.tex; pdf/cn/main_cn_jf.md.

Draft status: structurally complete working paper (theory + real PIT empirics + agenda). Acceptance at JF/RFS still requires deeper vintaged histories, logged outages, broader cross-section, and journal-format polishing—without dropping the formal apparatus again.

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